



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
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## **Unit 3 - Descriptive Statistics**

### **EXPLORATORY DATA ANALYSIS SOLVED PROBLEMS**

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## **1. MEAN, MEDIAN, MODE AND RANGE**

**Problem:** Find the mean, median, mode, and range for the dataset:

*12, 15, 18, 15, 20, 22, 15, 25*

**Step 1: Arrange the data in ascending order**

*12, 15, 15, 15, 18, 20, 22, 25*

Number of observations,  $n = 8$ .

**Step 2: Calculate the Mean**

$$\text{Mean} = (\Sigma x) / n$$

$$\text{Sum} = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

$$\text{Mean} = 142 / 8 = 17.75$$

**Step 3: Calculate the Median**

Since  $n = 8$  (even), the median is the average of the 4th and 5th values in the sorted data (15 and 18).

$$\text{Median} = (15 + 18) / 2 = 16.5$$

**Step 4: Calculate the Mode**

The value that occurs most frequently in the dataset is 15, which appears three times (all other values appear once).

$$\text{Mode} = 15$$

**Step 5: Calculate the Range**

$$\text{Range} = \text{Maximum} - \text{Minimum} = 25 - 12 = 13$$

**Final Answer: Mean = 17.75, Median = 16.5, Mode = 15, Range = 13**

## **2. FREQUENCY DISTRIBUTION**

**Problem:** Construct a frequency distribution for the dataset:

*2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4*

**Step 1: List the distinct values present in the data**

The distinct values in the dataset are: 2, 3, 4, 5, and 6.

**Step 2: Tally the occurrences of each value**

Counting how many times each value appears in the dataset of 12 observations:

➤ **Value 2:** appears 3 times (positions 1, 4, 8)

- **Value 3:** appears 3 times (positions 2, 6, 11)
- **Value 4:** appears 3 times (positions 3, 7, 12)
- **Value 5:** appears 2 times (positions 5, 10)
- **Value 6:** appears 1 time (position 9)

### Step 3: Construct the frequency table

| Value (x) | Frequency (f) |
|-----------|---------------|
| 2         | 3             |
| 3         | 3             |
| 4         | 3             |
| 5         | 2             |
| 6         | 1             |
| Total     | 12            |

### Step 4: Verify the total

The sum of all frequencies ( $3 + 3 + 3 + 2 + 1 = 12$ ) matches the total number of observations in the original dataset, confirming the frequency distribution is complete and correct.

**Final Answer: Frequency Distribution: 2→3, 3→3, 4→3, 5→2, 6→1 (Total = 12)**

## 3. QUARTILES (Q1, Q2, Q3) AND IQR

**Problem: Find Q1, Q2, Q3, and IQR for the dataset:**

*5, 8, 10, 12, 15, 18, 20, 22, 25, 30*

The data is already arranged in ascending order. Number of observations,  $n = 10$ .

### Step 1: Find Q2 (the Median)

Since  $n = 10$  (even), Q2 is the average of the 5th and 6th values (15 and 18).

$$Q2 = (15 + 18) / 2 = 16.5$$

### Step 2: Find Q1 (Lower Quartile)

Q1 is the median of the lower half of the data (the first 5 values):

*Lower half: 5, 8, 10, 12, 15*

*Q1 = 10 (the middle value of the lower half)*

**Step 3: Find Q3 (Upper Quartile)**

Q3 is the median of the upper half of the data (the last 5 values):

*Upper half: 18, 20, 22, 25, 30*

*$Q3 = 22$  (the middle value of the upper half)*

**Step 4: Calculate the Interquartile Range (IQR)**

$$IQR = Q3 - Q1 = 22 - 10 = 12$$

**Final Answer:  $Q1 = 10$ ,  $Q2 = 16.5$ ,  $Q3 = 22$ ,  $IQR = 12$**

**4. OUTLIER DETECTION – IQR METHOD**

**Problem: Detect outliers using the IQR method for the dataset:**

*10, 12, 13, 15, 16, 17, 18, 20, 45*

The data is already sorted. Number of observations,  $n = 9$ .

**Step 1: Find the Median ( $Q2$ )**

With  $n = 9$  (odd), the median is the 5th value.

$$Q2 = 16$$

**Step 2: Find  $Q1$** 

$Q1$  is the median of the lower half (values below the median): 10, 12, 13, 15.

$$Q1 = (12 + 13) / 2 = 12.5$$

**Step 3: Find  $Q3$** 

$Q3$  is the median of the upper half (values above the median): 17, 18, 20, 45.

$$Q3 = (18 + 20) / 2 = 19$$

**Step 4: Calculate the IQR**

$$IQR = Q3 - Q1 = 19 - 12.5 = 6.5$$

**Step 5: Calculate the Lower and Upper Bounds**

$$Lower\ Bound = Q1 - 1.5 \times IQR = 12.5 - 1.5(6.5) = 12.5 - 9.75 = 2.75$$

$$Upper\ Bound = Q3 + 1.5 \times IQR = 19 + 1.5(6.5) = 19 + 9.75 = 28.75$$

**Step 6: Identify Outliers**

Any value below 2.75 or above 28.75 is considered an outlier.

$$45 > 28.75$$

No value in the dataset falls below the lower bound of 2.75.

**Final Answer: 45 is an outlier. No lower-side outliers exist in this dataset.**

## **5. SKEWNESS (ASYMMETRY)**

**Problem: Determine the skewness of the dataset:**

*5, 6, 7, 7, 8, 8, 9, 20*

The data is already sorted. Number of observations,  $n = 8$ .

**Step 1: Calculate the Mean**

$$\text{Sum} = 5 + 6 + 7 + 7 + 8 + 8 + 9 + 20 = 70$$

$$\text{Mean} = 70 / 8 = 8.75$$

**Step 2: Calculate the Median**

Since  $n = 8$  (even), the median is the average of the 4th and 5th values (7 and 8).

$$\text{Median} = (7 + 8) / 2 = 7.5$$

**Step 3: Compare Mean and Median**

$$\text{Mean} = 8.75, \text{Median} = 7.5, \text{so Mean} > \text{Median}$$

**Step 4: Interpret the Result**

When the mean is greater than the median, the distribution is positively (right) skewed. In this dataset, the extreme value 20 pulls the mean toward the right while the median remains relatively unaffected, since the median depends only on the position of values rather than their magnitude.

**Final Answer: The distribution is positively (right) skewed, driven by the extreme value 20.**

## **6. UNIFORM DISTRIBUTION PROBABILITY**

**Problem: A uniformly distributed variable ranges from 10 to 50. Find  $P(20 < X < 35)$ .**

**Step 1: Identify the Distribution Limits**

$$a = 10, b = 50$$

**Step 2: Apply the Continuous Uniform Distribution Formula**

For a continuous uniform distribution, the probability that  $X$  falls between two values  $c$  and  $d$  is proportional to the length of that interval relative to the total range:

$$P(c < X < d) = (d - c) / (b - a)$$

**Step 3: Substitute the Given Values**

$$P(20 < X < 35) = (35 - 20) / (50 - 10) = 15 / 40$$

**Step 4: Simplify**

$$P(20 < X < 35) = 0.375$$

**Final Answer:**  $P(20 < X < 35) = 0.375$ , i.e., there is a 37.5% probability that X falls between 20 and 35.

## **7. Z-SCORE FOR A NORMAL DISTRIBUTION**

**Problem:** A normal distribution has mean 100 and standard deviation 15. Find the Z-score for  $X = 130$ .

**Step 1: Recall the Z-Score Formula**

$$Z = (X - \mu) / \sigma$$

where  $\mu$  is the population mean and  $\sigma$  is the population standard deviation.

**Step 2: Substitute the Given Values**

$$Z = (130 - 100) / 15 = 30 / 15$$

**Step 3: Simplify**

$$Z = 2.0$$

A Z-score of 2.0 means that  $X = 130$  lies exactly 2 standard deviations above the mean.

**Final Answer:**  $Z = 2.0$

## **8. PROBABILITY FROM Z-SCORE — $P(X < 130)$**

**Problem:** For the same normal distribution (mean = 100, SD = 15), find  $P(X < 130)$ .

**Step 1: Use the Z-Score Already Calculated**

From the previous problem,  $X = 130$  corresponds to:

$$Z = 2.0$$

**Step 2: Find the Cumulative Probability from the Standard Normal Table**

The standard normal (Z) table gives the cumulative probability  $P(Z < 2.0)$ , i.e., the area under the standard normal curve to the left of  $Z = 2.0$ :

$$P(Z < 2.0) = 0.9772$$

**Step 3: Relate Back to X**

Since  $Z = 2.0$  corresponds to  $X = 130$ , the cumulative probability  $P(X < 130)$  equals  $P(Z < 2.0)$ .

$$P(X < 130) = 0.9772$$

#### Step 4: Interpret the Result

This means approximately 97.72% of the values in this normal distribution are expected to fall below 130, and only about 2.28% of values are expected to exceed 130.

**Final Answer:**  $P(X < 130) \approx 0.9772$  (97.72%)

## 9. EFFECT OF OUTLIER TREATMENT ON MEAN AND MEDIAN

**Problem:** Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

To illustrate this concretely, the outlier dataset from Problem 4 is reused, since it contains a single clear extreme value (45):

$$10, 12, 13, 15, 16, 17, 18, 20, 45$$

#### Step 1: Calculate Mean and Median Before Treatment

Sum =  $10 + 12 + 13 + 15 + 16 + 17 + 18 + 20 + 45 = 166$ , with  $n = 9$ .

$$\text{Mean (before)} = 166 / 9 \approx 18.44$$

$$\text{Median (before)} = 16 \text{ (the 5th value in the sorted data)}$$

#### Step 2: Treatment Option A — Remove the Outlier

Removing 45 leaves the dataset: 10, 12, 13, 15, 16, 17, 18, 20 ( $n = 8$ ).

Sum =  $10 + 12 + 13 + 15 + 16 + 17 + 18 + 20 = 121$

$$\text{Mean (after removal)} = 121 / 8 = 15.125$$

$$\text{Median (after removal)} = (15 + 16) / 2 = 15.5$$

#### Step 3: Treatment Option B — Replace the Outlier with the Median

Replacing 45 with the pre-treatment median (16) gives: 10, 12, 13, 15, 16, 16, 17, 18, 20 ( $n = 9$ ).

Sum =  $10 + 12 + 13 + 15 + 16 + 16 + 17 + 18 + 20 = 137$

$$\text{Mean (after replacement)} = 137 / 9 \approx 15.22$$

$$\text{Median (after replacement)} = 16 \text{ (the 5th value, unchanged)}$$

#### Step 4: Compare and Interpret

| Measure | Before Treatment | After Removing Outlier | After Median Replacement |
|---------|------------------|------------------------|--------------------------|
| Mean    | 18.44            | 15.125                 | 15.22                    |



| Measure | Before Treatment | After Removing Outlier | After Median Replacement |
|---------|------------------|------------------------|--------------------------|
| Median  | 16               | 15.5                   | 16                       |

Before treatment, the mean (18.44) is noticeably higher than the median (16) because a single extreme value of 45 pulls the mean upward — the mean is sensitive to extreme values, while the median is not. After either treatment, the mean drops close to the median, showing a dataset that is far more representative of the "typical" values. This confirms the well-known property that the median is a more robust (outlier-resistant) measure of central tendency than the mean.

**Final Answer: Outlier treatment brings the mean much closer to the median (18.44 → ~15.1–15.2 vs. median ~15.5–16), confirming the median's robustness to extreme values.**

## **10. COMPLETE EDA ON A STUDENT-MARK DATASET**

**Problem: Perform a complete exploratory data analysis (EDA) on a small student-mark dataset and report central tendency, dispersion, distribution, outliers, and skewness.**

**Dataset — marks (out of 100) scored by 10 students in a test:**

*35, 40, 42, 45, 48, 50, 52, 55, 58, 92*

The data is already arranged in ascending order. Number of observations,  $n = 10$ .

### **Step 1: Central Tendency**

$$\text{Sum} = 35 + 40 + 42 + 45 + 48 + 50 + 52 + 55 + 58 + 92 = 517$$

$$\text{Mean} = 517 / 10 = 51.7$$

Since  $n = 10$  (even), the median is the average of the 5th and 6th values (48 and 50).

$$\text{Median} = (48 + 50) / 2 = 49$$

All ten values in the dataset are distinct, so there is no repeating value.

$$\text{Mode} = \text{None (no value repeats)}$$

### **Step 2: Dispersion (Range, Variance, Standard Deviation)**

$$\text{Range} = \text{Maximum} - \text{Minimum} = 92 - 35 = 57$$

The (population) variance is calculated using the mean of 51.7. The squared deviation of each value from the mean is computed and averaged:

| Mark (x) | (x – Mean) | (x – Mean) <sup>2</sup> |
|----------|------------|-------------------------|
| 35       | –16.7      | 278.89                  |
| 40       | –11.7      | 136.89                  |

| Mark (x) | (x – Mean) | (x – Mean) <sup>2</sup> |
|----------|------------|-------------------------|
| 42       | –9.7       | 94.09                   |
| 45       | –6.7       | 44.89                   |
| 48       | –3.7       | 13.69                   |
| 50       | –1.7       | 2.89                    |
| 52       | 0.3        | 0.09                    |
| 55       | 3.3        | 10.89                   |
| 58       | 6.3        | 39.69                   |
| 92       | 40.3       | 1624.09                 |

Sum of squared deviations =  $278.89 + 136.89 + 94.09 + 44.89 + 13.69 + 2.89 + 0.09 + 10.89 + 39.69 + 1624.09 = 2246.10$

$$\text{Population Variance } (\sigma^2) = 2246.10 / 10 = 224.61$$

$$\text{Population Standard Deviation } (\sigma) = \sqrt{224.61} \approx 14.99$$

(Using the sample formula with  $n - 1 = 9$  in the denominator instead gives a sample variance of  $\approx 249.57$  and sample SD  $\approx 15.80$ .)

### Step 3: Quartiles and IQR

Lower half (first 5 values): 35, 40, 42, 45, 48 → median = 42, so  $Q1 = 42$ .

Upper half (last 5 values): 50, 52, 55, 58, 92 → median = 55, so  $Q3 = 55$ .

$$IQR = Q3 - Q1 = 55 - 42 = 13$$

### Step 4: Outlier Detection (IQR Method)

$$\text{Lower Bound} = Q1 - 1.5 \times IQR = 42 - 19.5 = 22.5$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR = 55 + 19.5 = 74.5$$

Any mark below 22.5 or above 74.5 is flagged as an outlier.

$$92 > 74.5$$

**Final Answer: 92 is a statistical outlier — it likely represents an exceptionally high-performing student rather than a data entry error, and should be investigated rather than deleted automatically.**

### Step 5: Skewness (Distribution Shape)

$$\text{Mean} = 51.7, \text{Median} = 49, \text{so Mean} > \text{Median}$$

Since the mean exceeds the median, the distribution of marks is positively (right) skewed. This is consistent with the presence of the high outlier (92), which pulls the mean upward while the bulk of the class scored in the 35–58 range.

#### Step 6: Summary of the EDA

| Aspect                        | Result                                  |
|-------------------------------|-----------------------------------------|
| Mean                          | 51.7                                    |
| Median                        | 49                                      |
| Mode                          | None (all values distinct)              |
| Range                         | 57                                      |
| Population Standard Deviation | $\approx 14.99$                         |
| Q1 / Q3                       | 42 / 55                                 |
| IQR                           | 13                                      |
| Outlier(s)                    | 92 (above the upper bound of 74.5)      |
| Skewness                      | Positive / right-skewed (Mean > Median) |

Overall interpretation: the class performance is broadly clustered between 35 and 58 marks, with one exceptional score of 92 that behaves as a statistical outlier and is responsible for the right-skew of the distribution. A fair summary of "typical" performance should favour the median (49) over the mean (51.7), since the mean is distorted upward by the single high score.